

DSA0502 – Query Processing for Data Science	
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LAB PROGRAMS 9 AND 10

Program 9

Question: Create a Pivot table and find the total sale amount region wise, manager wise, salesman wise.

Aim: To implement the given Pandas program successfully.

Dataset :

%%writefile sales.csv

Item,Region,Manager,Salesman,Units_Sold,Sale_Value

Laptop,North,Raj,Amit,10,500000

Laptop,South,Priya,Ravi,12,600000

Laptop,East,Raj,Sneha,8,400000

Laptop,West,John,Kiran,15,750000

Mouse,North,Priya,Amit,20,20000

Mouse,South,John,Ravi,25,25000

Mouse,East,Raj,Sneha,18,18000

Mouse,West,Priya,Kiran,22,22000

Keyboard,North,Raj,Amit,15,45000

Keyboard,South,John,Ravi,20,60000

Keyboard,East,Priya,Sneha,16,48000

Keyboard,West,Raj,Kiran,14,42000

Monitor,North,John,Amit,5,75000

Monitor,South,Priya,Ravi,6,90000

Monitor,East,Raj,Sneha,4,60000

Monitor,West,John,Kiran,7,105000

Code

```
import pandas as pd
df=pd.read_csv('sales.csv')
print(pd.pivot_table(df,index=['Region','Manager','Salesman'],values='Sale_Value',aggfunc='sum'))
```

Output

Region/Manager/Salesman wise sales.

...	Region	Manager	Salesman	Sale_Value
	East	Priya	Sneha	48000
		Raj	Sneha	478000
	North	John	Amit	75000
		Priya	Amit	20000
		Raj	Amit	545000
	South	John	Ravi	85000
		Priya	Ravi	690000
	West	John	Kiran	855000
		Priya	Kiran	22000
		Raj	Kiran	42000

Result

Thus, the program was executed successfully and the expected output was obtained.

Program 10

Question: Create a dataframe of ten rows and four columns with random values. Highlight negative numbers red and positive numbers black.

Aim: To implement the given Pandas program successfully.

Dataset :

Random DataFrame (10 x 4) generated using NumPy.

Code

```
import pandas as pd, numpy as np
df=pd.DataFrame(np.random.randn(10,4))
df.style.applymap(lambda x:'color:red' if x<0 else 'color:black')
```

Output

Styled dataframe.

```
... /tmp/ipykernel_1262/440386146.py:3: FutureWarning: Styler.applymap has been deprecated. Use Styler.map in
df.style.applymap(lambda x:'color:red' if x<0 else 'color:white')
```

	0	1	2	3
0	0.310770	-1.525460	0.179680	0.331170
1	0.975806	1.099792	-0.535110	-1.535104
2	-2.561741	2.602457	1.783820	-2.529423
3	0.995965	-1.077417	-0.395752	-1.121889
4	0.503402	0.525621	1.013825	-0.657647
5	-0.318817	-1.256582	0.158574	1.575107
6	0.636794	0.535710	-0.790281	-0.823887
7	1.420267	-0.156862	-0.737551	1.364297
8	-1.043283	-0.241563	0.710664	0.793920
9	-0.733827	0.965778	0.080298	-0.149534

Result

Thus, the program was executed successfully and the expected output was obtained.